Alvin I. Ugo-Mgbemene

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Summary

Product Development Engineer specializing in silicon validation, qualification, yield ramp, and failure analysis in high-volume semiconductor manufacturing. Experienced driving NPI through production release while improving yield, reducing scrap, analyzing wafer- and package-level failures, and building scalable automation for faster characterization and release readiness.

Education

University of Pennsylvania — Philadelphia, PA
M.S.E. Data Science — Starting Spring 2026

Texas A&M University — College Station, TX
B.S.E. Electrical & Computer Engineering, magna cum laude — GPA: 3.73/4.00

Technical Skills

Languages: Python, SQL, C++
Systems: Linux, Docker, Git
Data & Analytics: ETL pipelines, pandas, NumPy, DBSCAN, anomaly detection, regression workflows
Statistical & Validation: JMP, Power BI, Spotfire, yield modeling, failure correlation, validation analysis
Semiconductor PDE: NPI ramp, silicon validation, qualification, wafer/package failure analysis, characterization, DOE, test/debug, device physics

Experience

Texas Instruments — Sherman, TX
Product Development Engineer — June 2023 – Present

Impact: 14x faster NPI planning — 20% flagship yield improvement — 10% additional yield improvement — 30% scrap reduction

- Led silicon validation and qualification readiness for high-volume semiconductor products from NPI through production ramp in a startup fab environment.
- Owned execution across NPI, ramp, and qualification phases, ensuring manufacturability, configuration integrity, and alignment with silicon, package, and reliability requirements.
- Developed and deployed a Python application converting silicon design constraints into executable fab device flows, reducing NPI planning cycles from 2 weeks to 1 day.
- Supported silicon and package-level qualification activities, including HTOL, electromigration, and burn-in, ensuring flows met reliability and release criteria.
- Drove failure analysis across qualification and production excursions by correlating yield, defect density, inline metrology, test fallout, and process signals to isolate root causes.
- Diagnosed tester-related marginality affecting validation accuracy, improving measurement robustness and eliminating false failure signatures.
- Partnered with test, process, product, and manufacturing teams to validate test conditions, analyze failure modes, and improve characterization accuracy.
- Accelerated yield ramp on flagship products (+20%) and additional lines (+10%) through structured validation, regression control, characterization analysis, and failure-driven optimization.
- Reduced scrap by about 30% by enforcing manufacturability constraints and eliminating invalid product configurations before release.

Procter & Gamble — Cincinnati, OH
Engineering Intern — May 2022 – Aug 2022

- Built ETL and statistical modeling pipelines to forecast resource usage and detect production bottlenecks.
- Delivered analytics supporting decisions that contributed to $1.5M in projected annual savings.

Texas A&M University — College Station, TX
Undergraduate Research Assistant — Jan 2022 – May 2022

- Developed distributed simulation tools using Python and ZeroMQ for real-time system communication.
- Improved control and signal-processing throughput by approximately 25%.

Leadership

Process Analytics Training Lead, Texas Instruments — 2025 – Present
Led training on statistical validation, anomaly detection, yield correlation, and failure analysis.

TI Data, Systems, and Software Symposium (DS3) — 2024 – Present
Reviewer for internal technical submissions evaluating analytical rigor and methodology.